Optical Manipulation of Cells

Julian Cheng, M. Arifur Rahman and Aaron T. Ohta

Abstract Single-cell analysis can reveal cell behaviors that are unobservable by traditional bulk measurements. The analysis of single cells is made possible by technology that enables the addressing and manipulation of microscale objects. Optical manipulators are capable of micro- and nano-manipulation and are attractive for single-cell procedures, due to their inherent flexibility and adaptability. This chapter reviews major types of optical manipulation techniques. Optical gradient and scattering forces, used in optical tweezers, holographic optical tweezers, and other optical traps, are introduced first. Next, optically controlled electrokinetic forces are discussed, with a focus on optically induced dielectrophoresis. Optically induced thermal effects are also introduced, including thermophoresis and thermocapillary force. Specific examples of applications of these techniques for procedures related to single-cell analysis are also covered, including cell culturing, cell sorting, cell surgery, and measurements of single cells.

Keywords Single-cell analysis · Cell manipulation · Optical manipulation · Optical tweezers · Holographic optical tweezers · Optically induced dielectrophoresis · Opto-thermocapillary flow · Cell culturing · Cell sorting · Cell trapping · Molecular delivery · Cell lysis

1 Introduction

Cellular behavior has traditionally been studied by observing the response of a population of cells. While this has led to breakthroughs in understanding biological processes, there is the risk of losing information on the behavior of rare cells

J. Cheng · M.A. Rahman · A.T. Ohta (✉)
Department of Electrical Engineering, University of Hawaii, Honolulu, HI, USA
e-mail: aohta@hawaii.edu

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or cells with responses that deviate substantially from the mean (Di Carlo and Lee 2006). Thus, as the capability to handle individual cells arose, measurements at the cellular level have yielded new insights (Lindström and Andersson-Svahn 2010; Wang and Bodovitz 2010). In order to study single cells, desirable capabilities include

- Cell culturing
- Cell separation and sorting
- Cell trapping, isolation, and transport
- Operations on cells, such as cell lysis and molecular delivery
- Measurements of cellular properties, such as mechanical stiffness and mass

All of these capabilities can be performed or assisted by optical systems. This includes systems that use optical forces, such as optical gradient and scattering forces, as well as systems that use optical control of other types of forces, such as optically induced electrokinetic forces, and optically induced thermal forces. These types of optical systems will be described in this chapter, with an emphasis on their applications for the manipulation of single cells.

An important advantage of optical systems is their adaptability. An optical system is capable of being reconfigured on the fly, in contrast to mechanical micromanipulation systems. The optical patterns used to realize cell manipulation can be dynamically adjusted to switch between various functions, such as trapping and sorting, or to optimize the system for a specific need or application. Furthermore, optical patterns can be programmed or made part of feedback control systems, making it possible to automate complex operations. Optical manipulation is also noncontact, which can be gentler on fragile objects such as mammalian cells.

For cell manipulation applications, optical systems also have the advantage that it is straightforward to integrate optical manipulation with microfluidic structures by using transparent materials or optical waveguides. Furthermore, the use of optical manipulation within a microfluidic device adds functionality, which means that the mechanical microfluidic structures can be relatively simple. This results in reduced fabrication costs for the microfluidic devices, which are often disposable to reduce sample cross-contamination.

When using optical manipulation systems for cell manipulation, there are some limitations that arise. Some types of optical manipulation require highly intense light, which can affect the cells under manipulation. However, this is not true for all of the optical manipulation methods described in this chapter, especially those that do not rely on optical gradient forces.

The working area of optical manipulators is determined by the optical setup. Some types of optical manipulators have working areas of less than a square millimeter, while others can operate over hundreds of square millimeters. It may also be possible to mitigate some of the effects of a small working area by translating a microfluidic device in relation to the working area of the optical manipulator.

Optical systems typically require skilled technicians to set them up, and are relatively expensive. Commercial optical systems address this with preassembled optical components, but have increased cost and space requirements, and still

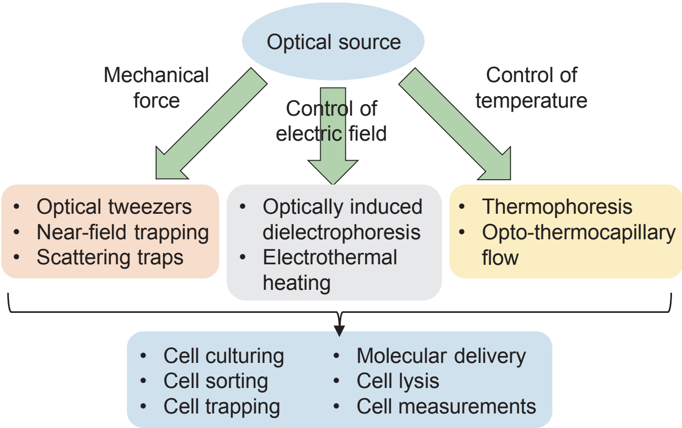

Fig. 1 Various methods of optical manipulation discussed in this chapter, and their applicability to single-cell analysis functions. Direct optical forces, where optical energy is converted to a mechanical force, are discussed, along with optically controlled manipulation mechanisms, including optical control of electric fields and temperatures

usually require a technician for the initial setup. Some optical manipulation systems use lower cost, compact components to address these limitations.

This chapter will describe various types of optical systems that are suitable for the manipulation of cells, include those that use direct optical forces, such as optical tweezers, and those that use optical control of other forces, such as optically controlled electrokinetics and optothermal manipulation (Fig. 1). The optical manipulation of cells using the various optical systems will then be discussed in more detail, organized by the capabilities mentioned at the beginning of this introduction.

2 Direct Optical Forces

Direct optical forces convert photon energy to mechanical force, resulting in optical gradient forces and scattering forces. These forces have been harnessed to create optical tweezers, near-field optical traps, and other types of traps.

2.1 Optical Tweezers

Optical tweezers are optical traps that can generate forces on the order of 0.1–100 piconewtons on micro- to nanoscale particles (Ashkin et al. 1986). Optical tweezers use the optical gradient force produced by a single tightly focused laser beam,

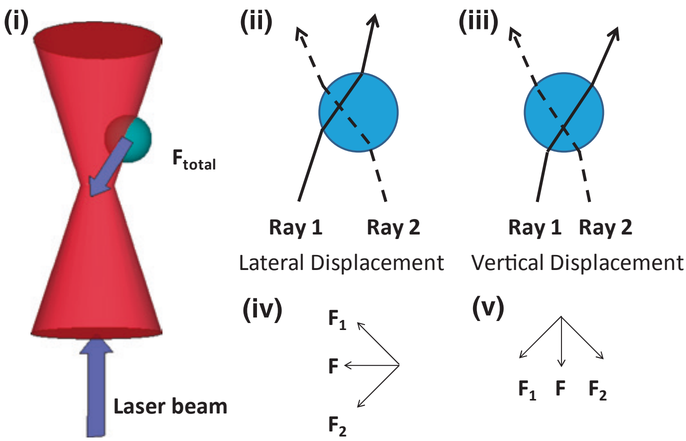

Fig. 2 Diagrams of a particle being trapped by optical tweezers. **i** The particle undergoes lateral and vertical displacement due to the gradient force and is attracted toward the focal point of the laser. **ii** A particle offset laterally from the laser focal point will move left toward the focal point of the beam. **iii** A particle offset vertically from the laser focal point will move down toward to the focal point of the beam. **iv** The free-body force diagram corresponding to the particle in **ii**. **v** The free-body force diagram corresponding to the particle in **iii**

which results from the refraction of light as it passes through a particle. Particles with an index of refraction greater than that of the surrounding media are attracted to the focal point, the region with the highest light intensity. If the laser is not focused, the gradient force only becomes significant in the transverse plane, confining the particle laterally, but propelling the particle along the laser beam by radiation pressure (Ashkin 1970). A tightly focused laser beam generates enough gradient force along the beam direction to counteract the radiation pressure, stably trapping a particle in space (Fig. 2). The difference in index of refraction between the particle and the surrounding medium, along with the optical power of the trap, contribute to how strongly the particle is trapped. The force exerted by an optical tweezers trap can be described by Hooke's Law according to $F = -kx$, where F is the force on an object, and x is the displacement of the object from the middle of the trap. The spring constant, k, depends on the properties of the laser, and describes how "stiff" the trap is.

2.1.1 Optical Tweezers for the Manipulation of Biological Materials

Optical tweezers have been used for the manipulation of many biological materials, including DNA strands (Waleed et al. 2013), proteins in DNA (van Loenhout et al. 2013), nucleic acid motor enzymes (Yin et al. 1995), and tobacco mosaic viruses (Ashkin and Dziedzic 1987). Optical tweezers have also been used to manipulate cells, including *E. coli* bacteria (Ashkin and Dziedzic 1987), plant cells (Ashkin et al. 1987), algal cells (Tanaka et al. 2008), yeast cells (Landenberger et al. 2012), live sperm cells (Chen et al. 2011), red blood cells (Constable et al. 1993), and embryonic stem cells (Wang et al. 2011). When manipulating biological materials, optical tweezers are usually combined with microfluidic systems, which can be used to provide the appropriate media. The optical tweezers functionality can then be used to supplement or complement the microfluidic system. For example, optical tweezers can be used to move cells into different sets of media (Enger et al. 2004).

When using optical tweezers, the laser energy absorbed by the biological material under manipulation should be considered. The high optical intensities in an optical trap (megawatts per cm^2) can cause photodamage (Dholakia and Reece 2006). The damage threshold varies for different types of cells, and is also dependent on the amount of absorption, which in turn is a function of laser wavelength. *E. coli* exhibited maximum photodamage at wavelengths of 870 and 930 nm, while minimum photodamage was seen at 830 and 970 nm (Neuman et al. 1999). Chinese ovary cells are also affected by laser energy, as quantified by measurements of their cloning efficiency: cloning efficiencies were greater than 75 % for wavelengths between 950 and 990 nm when the cell receives an optical energy dosage of approximately 9×10^9 J/cm^2 (Liang et al. 1996). Therefore, careful consideration must be made regarding the laser chosen for an optical tweezers system, especially when manipulating biological materials.

One way to reduce the possibility of photodamage is through the indirect manipulation of cells using optical tweezers. Microparticles are attached to the cells under manipulation, and optical tweezers are used to trap and position the microparticles, thus moving the attached cells (Arai et al. 2003). This avoids direct illumination of the cells, reducing photodamage. The use of these microparticles, or microtools, slightly reduces the flexibility of optical tweezers, since microtools need to be added to the working area, and cells need to be attached to the microtools. This has been addressed through the in situ creation of microtools using photopolymerizing polymers (Maruyama et al. 2005) or temperature-sensitive polymers (Ichikawa et al. 2005). The use of microtools also can add functionality to optical tweezers systems, including using the microtools as a pH sensor (Maruyama et al. 2007).

2.1.2 Multiple Optical Tweezers Traps

While single-beam force gradients have various applications in both sorting and manipulating both particles and cells, it is also important to consider the use of multiple optical traps in one setup. Having multiple optical traps would further expand the usefulness of optical tweezers as a large number of particles could be trapped and manipulated at once, enabling increased throughput and both parallel and cooperative manipulation.

Multiple-beam force gradients can be generated by splitting a single laser beam. This can be done by time-sharing, where a single beam is directed to multiple locations within a small period of time. This allows for multiple optical traps to be controlled, as long as the trap can prevent cell under observation from escaping. Time-sharing can be achieved by acousto-optical deflectors (AODs), although the diffraction efficiency is only about 80 % (Visscher et al. 1996), reducing the power available for trapping. An alternative is electro-optic deflectors (EOD), which compared to AODs, have less optical loss (Valentine et al. 2008). However, this method is limited to smaller deflection angles, thus confining particles to a smaller workspace. Another option is the use of scanning mirrors, which can operate at up to 1 kHz (Arai et al. 2004; Tanaka et al. 2008). However, like all time-sharing methods, the number of optical traps that can be created is limited by the minimum amount of time that each trap needs to be active.

2.2 Holographic Optical Tweezers

Holographic optical tweezers (HOT) takes a different approach to the creation of multiple optical traps (Fig. 3). In an HOT system, a diffraction pattern splits a single laser beam into several different beams, allowing for the control of multiple optical traps (Dufresne et al. 2001). Like single-beam optical tweezers, HOT are capable of three-dimensional manipulation (Grier and Roichman 2006; Lee and Grier 2007).

The spatial locations of the HOT traps can be reconfigured in real time by using a spatial light modulator (SLM), which uses computer-generated holograms to dynamically split an incident laser beam. The holograms displayed on the SLM can be easily changed using a computer, adjusting the optical trap positions (Curtis et al. 2002). HOT systems have been used for the manipulation of a variety of particles and cells, and have made it possible to create patterns of groups of cells. For example, HOT systems have manipulated 5-μm-diameter mouse embryonic stem cells into structures such as lines and curves (Leach et al. 2009).

An issue with multiple HOT traps is additional traps can be accidentally generated in undesired locations, creating "ghost traps" (Bianchi and Di Leonardo 2010), especially if a complex or symmetric pattern is used. Optical aberrations can also affect the effectiveness of an optical trap, causing the trap to be less focused. In addition, creating a homogeneous distribution of optical power among all the optical traps in an HOT system is not straightforward. In most cases, iterative algorithms (Di Leonardo et al. 2007) can be used to solve these issues. These algorithms add to the computing time, but most consumer graphic cards and GPU-based systems can generate the appropriate holograms to correct for intrinsic and induced optical aberrations before the Brownian motion of the particle occurs (Bianci and Di Leonardo 2010; Bowman and Padgett 2013; Bowman et al. 2014; Martín-Badosa et al. 2007). Algorithms such as an iterative Fourier transform can eliminate the majority of ghost spots and reduce the intensity of remaining ghost spots to less than 5 % of the mean intensity, making it difficult for particles to be trapped in the wrong locations (Hesseling et al. 2011). Another iterative algorithm uses a Shack–Hartmann array to help correct aberrations before the hologram is projected

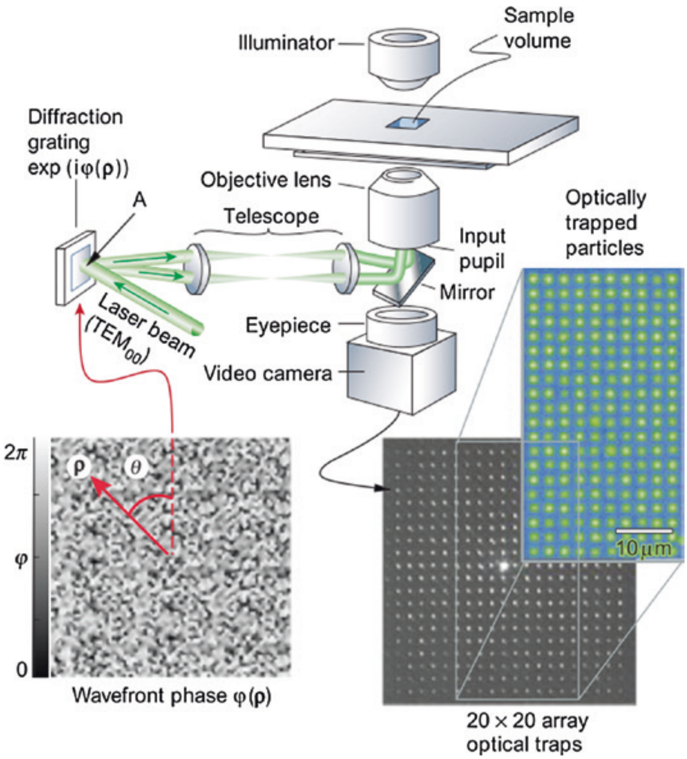

Fig. 3 A holographic optical tweezers system uses a diffraction pattern to split a single laser beam into multiple optical tweezers traps. Reprinted from Grier (2003) with permission from MacMillian Publishers Limited

to the SLM (Bowman et al. 2010), reducing possible errors while trapping a particle. Through the use of iterative algorithms, multiple HOT traps can be made more effective. Red Tweezers, a HOT control program developed by the University of Cambridge, is an example of software that utilizes iterative algorithms (Bowman et al. 2014). The program, which runs in LabVIEW (National Instruments), is able to generate holograms at high speed and deliver the data to an SLM.

2.3 *Other Optical Traps*

2.3.1 Near-Field Optical Traps

Near-field optical traps use evanescent optical fields to generate optical gradient forces capable of trapping particles. It is particularly effective for trapping nanoscale particles. One way to realize near-field optical trapping is near the surface of optical waveguides (Yang et al. 2009; Soltani et al. 2014) or optical resonators (Lin et al. 2010; Mandal et al. 2010; Hu et al. 2010). Near-field optical trapping can also be achieved through the use of plasmonic nanostructures, which have been shown to be capable of trapping biological materials, such as yeast cells (Huang et al. 2009). However, one concern when working with biological materials is the high temperature near the plasmonic structures.

2.3.2 Scattering Forces and Other Forces

Optical scattering forces can also be used for cell manipulation instead of optical gradient forces. Unlike an optical tweezers trap, where cells are trapped in three dimensions, optical scattering forces can push cells along the direction of laser beam propagation. Two or more laser beams can use scattering forces to stably trap objects in three dimensions, if the laser beams are directed toward each other (Constable et al. 1993; Yuan et al. 2008; Jess et al. 2006; Lincoln et al. 2007). Particles are attracted toward the dual beams by their gradient forces, and the scattering forces of each beam dictate the position of the particle relative to the laser sources. Assuming the laser powers are equal, the stable trapping position will be in the center of the laser beams, at a point midway between the two lasers. Unlike a single-beam optical trap, a dual-beam optical trap does not require high numerical aperture optics or a tightly focused beam; this means less photodamage on cells or biomolecules (Jonáš and Zemánek 2008). However, scaling up the number of discrete optical traps using dual beams is more complex than with a single-beam optical tweezers trap.

Optical scattering forces can be used to extract specific cells out of microwells for further analysis or disposal (Kovac and Voldman 2007), or for continuous-flow microfluidic cell sorters (Lee et al. 2013). Optical interference patterns and other sculpted optical landscapes can also be used to direct the trajectories of cells for sorting (MacDonald et al. 2003; Jákl et al. 2014). These sorting methods are explained in more detail in Sect. 5.2.

Another interesting method for cell manipulation is the use of a "tractor beam" (Brzobohatý et al. 2013). Unlike optical tweezers, tractor beams use a pulling force to transport particles against the direction of beam propagation, while gradient force is only used to keep the particle in the plane transverse to the propagation direction. This pulling force is achieved with two interfering plane waves intersecting at an angle, and depends on the beam polarization. Notably, this method should be compatible with biological particles, including cells, which have similar properties to the polystyrene beads used in these experiments.

3 Optically Induced Dielectrophoresis (ODEP)

The contactless manipulation of cells can also be accomplished using a low-frequency analogue to optical tweezers, dielectrophoresis (DEP) (Gagnon 2011; Pethig 2013). An applied electric field can induce a dipole across a cell;

if the electric field is nonuniform, each end of the dipole experiences a different Coulombic force, resulting in a net dielectrophoretic force. The nonuniform electric fields necessary for DEP can be created by microfabricated electrodes (Jubery et al. 2014; Cetin and Li 2011), but DEP can also be optically controlled using optoelectronic tweezers, or optically induced dielectrophoresis (ODEP) (Chiou et al. 2005; Wu 2011). Many cell types have been manipulated by ODEP, including bacteria (Chiou et al. 2004), many types of mammalian cells (Ohta et al. 2007a, b; Hwang et al. 2008a; Neale et al. 2009a), and protozoa (Choi et al. 2008). ODEP has also been used for trapping DNA (Lin et al. 2009) and cells encapsulated in hydrogel beads (Lin et al. 2012a).

In ODEP, a photosensitive material is used to create the nonuniform electric field necessary for DEP using optical patterns (Fig. 4i). This is done by using light to change the electrical impedance of the photosensitive material. In turn, this changes the strength of the electric field in a liquid layer abutting the photosensitive material, and the resulting non-uniform electric field generates a DEP force.

As ODEP is optically controlled, this technique retains the flexibility and dynamic control of direct optical manipulation. However, since the optical energy is not directly converted to the trapping force, the optical requirements can be relaxed. Significantly lower optical intensities can be used to control ODEP as compared to optical gradient traps; typically, intensities from 10 to 1000 mW/cm^2 are used for ODEP. The optical sources can be coherent (Neale et al. 2007) or noncoherent, such as mercury lamps (Ohta et al. 2007a) or LEDs (Chiou et al. 2005; Zarowna-Dabrowska et al. 2011). Light patterns can be formed from a single light source using micromirror arrays and other spatial light modulators, or by simply taking the output of a computer projector (Ohta et al. 2007a) or LCD (Choi et al. 2007; Hwang et al. 2008a). Computer projectors and LCDs have the advantage that the light source and the mechanism for patterning the light are integrated, facilitating the setup of the ODEP system, and enabling a relatively compact footprint.

3.1 Operation of a Typical ODEP System

A typical ODEP device has a transparent planar electrode made of indium-tin oxide (ITO) on glass and a photosensitive electrode (Fig. 4i). The two electrodes house a fluidic chamber between them, which contains the cells under manipulation. An electric field is created in the fluidic chamber by applying an AC voltage across the electrodes, typically at a frequency in the range of tens to hundreds of kilohertz.

Optical patterns are focused onto the photosensitive electrode to modulate the electric field. Under dark or ambient lighting, the photosensitive electrode has a low conductivity, and its impedance, Z_{PC}, is larger than the impedance of the liquid layer, Z_L. This means that the majority of the applied AC voltage is across the photosensitive layer. However, when light is focused on the photosensitive electrode, Z_{PC} is lowered enough that a significant voltage is dropped across the liquid

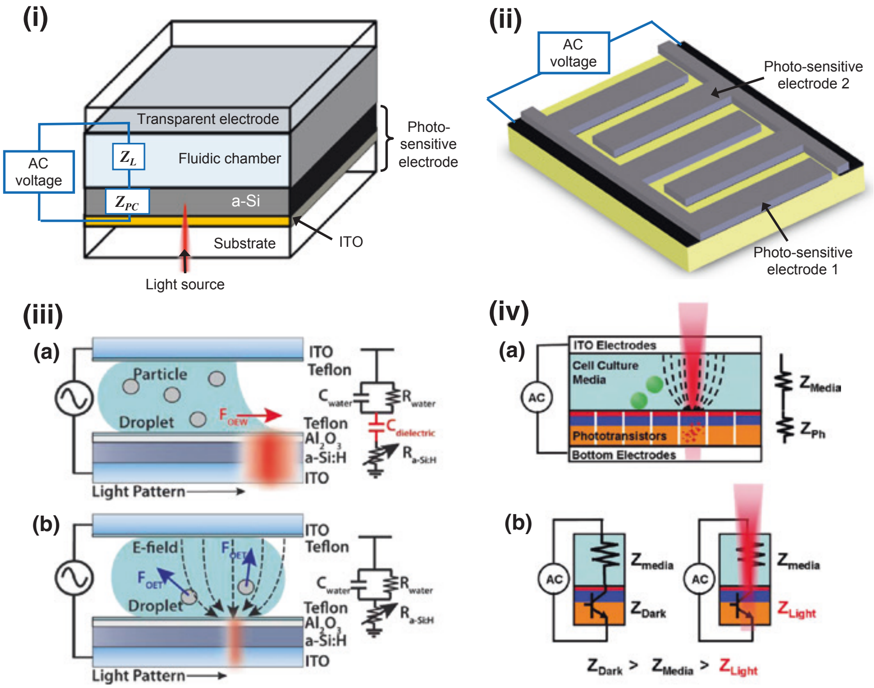

Fig. 4 Various configurations of optically induced dielectrophoresis devices. **i** The standard device for optically induced dielectrophoresis. It consists of a fluidic chamber bounded by a transparent electrode and a photosensitive electrode. The photosensitive electrode is usually amorphous silicon (a-Si) on indium-tin oxide (ITO). **ii** Lateral-field optoelectronic tweezers create electric fields that are primarily parallel to the surface, and consolidates the ODEP electrodes to one surface. **iii** Optoelectrowetting and optically induced dielectrophoresis can be realized on the same device by changing the frequency of the electric field. Reproduced from Valley et al. (2011) with permission from the Royal Society of Chemistry. **iv** Phototransistor-based optoelectronic tweezers have increased conductivity under illumination, enabling operation in cell culture media and other physiological solutions. Reproduced from Hsu et al. (2010) with permission from the Royal Society of Chemistry

layer. This switching between a normally low-conductivity state and high-conductivity illuminated state enables the creation of optically defined electrodes on the photosensitive electrode (Ohta et al. 2007a). Thus, the light patterns shape the electric field landscape, creating electric field gradients in the liquid layer between the illuminated and dark areas of the photosensitive electrode, generating ODEP force, which, for spherical particles such as cells, is given by the following equation (Jones 1995):

$$F_{\mathrm{DEP}} = 2\pi r^3 \varepsilon_m \mathrm{Re}[K(\omega)] \nabla E_{\mathrm{rms}}^2 \tag{1}$$

where r is the particle radius, ε_m is the permittivity of the medium surrounding the particles, $K(\omega)$ is the Clausius–Mossotti factor, and E is the electric field.

The Clausius–Mossotti factor is a complex number that depends upon the frequency of the applied electric field and the properties of the particle and the surrounding medium, as given by (Jones 1995):

$$K(\omega) = \frac{\varepsilon_p^* - \varepsilon_m^*}{\varepsilon_p^* + 2\varepsilon_m^*} \tag{2}$$

where ε_p^* and ε_m^* are the complex permittivities of the particle and medium, respectively. These are given by (Jones 1995):

$$\varepsilon_{p,m}^* = \varepsilon_{p,m} - j\frac{\sigma_{p,m}}{\omega} \tag{3}$$

where the permittivity, ε, and conductivity, σ, are for either the particle, p, or medium, m. The angular frequency of the electric field is represented by ω. Thus, the ODEP force depends not only on the volume of the particles, but also on the difference in the electrical properties of the particles and the surrounding media.

The most popular photosensitive electrode for ODEP consists of a layer of intrinsic amorphous silicon (a-Si) deposited on a layer of ITO, but other materials can also be used. Cadmium sulfide is another photoconductive semiconductor that has been used for ODEP (Higuchi et al. 2008), with the advantage that its chemical deposition is more inexpensive than the plasma-enhanced chemical vapor deposition used for a-Si. Organic photoconductors and polymers can simplify fabrication even more, such as spin-coated layers of titanium oxide phthalocyanine (Yang et al. 2011, 2013) or a blend of regioregular poly (3-hexylthiophene) and phenyl-C61-butyric acid methyl ester (Wang et al. 2010a). These materials are also compatible with flexible substrates (Lin et al. 2012b). Another photosensitive material, lithium niobate, enables the retention of the electric field patterns even after the illumination is removed, although it does take longer to switch the conductivity of this material between the light and dark states (60 s to a few minutes) (Glaesner et al. 2012). Other materials can also be added to the traditional a-Si layer; nanoparticles can be tethered to a lipid bilayer on the a-Si surface of the ODEP device, ensuring that these particles experience the maximum possible ODEP force (Ota et al. 2013).

Variants of the typical ODEP device have been created to add more functionality. To allow the easier integration of ODEP with polydimethylsiloxane (PDMS) microfluidic channels, the transparent planar electrode made of ITO-coated-glass was replaced with a different transparent structure consisting of carbon nanotubes embedded in PDMS (Huang et al. 2013a). It is also possible to facilitate the integration of ODEP with microchannels and other microfabricated structures by placing both ODEP electrodes on a single surface. This was accomplished with an interdigitated array of a-Si-coated electrodes (Fig. 4ii) (Ohta et al. 2007b). Placing the ODEP electrodes on a single surface has another effect: the electric field is primarily parallel to the plane of the electrodes, in contrast to the electric field that is normal to the electrodes in the typical ODEP device. Thus, in this variant of the ODEP device, anisotropic particles (Tien et al. 2009; Ohta et al. 2007c, 2008;

Neale et al. 2009b) align their long axis parallel to the surface of the substrate, which contrasts with the standard ODEP device (Jamshidi et al. 2008). Besides affecting the orientation of anisotropic particles, the single-side ODEP architecture allowed the integration of ODEP force for cell trapping and transport with electrowetting-on-dielectric (EWOD) for droplet manipulation (Shah et al. 2009).

Droplet manipulation can also be achieved by adding a dielectric layer and hydrophobic coating over the photosensitive electrode (Valley et al. 2011) (Fig. 4iii). At AC voltages around 10 kHz, this device acts as an optically controlled EWOD device, and the light patterns control the movement of droplets (Fig. 4iii-a). When the frequency of the applied voltage is increased to around 200 kHz, the device acts as a standard ODEP device, and is capable of moving particles within the droplets using DEP force (Fig. 4iii-b).

3.2 Limitations of ODEP

There are several limitations of the standard ODEP system with respect to cellular manipulation. The first concern is throughput; in optical manipulation, this is typically limited by the area over which the optical patterns can be generated. This is less of an issue with ODEP as compared to optical tweezers, as the optical intensity and focusing requirements are more relaxed. Instead, throughput in a conventional ODEP system is limited by the system used to visualize the cells under manipulation, which is usually a microscope with a field-of-view a few square millimeters. Lens-free holographic imaging can greatly increase the area over which cells and other particles can be imaged (Mudanyali et al. 2010), and coupled with ODEP, can improve the throughput (Huang et al. 2013b). The working area of 240 mm^2 reported by Huang et al. is the largest effective area of any optical manipulator to date.

Although manipulation throughput can be higher using ODEP compared to optical tweezers, especially with larger particles such as cells, ODEP does have issues associated with the electrical conductivity of the media containing the cells under manipulation. Many types of physiological media, including phosphate-buffered saline and commonly used cell culture media, have relatively high concentrations of ions, resulting in high electrical conductivities (typically 1–2 S/m). This presents a problem for the photosensitive electrodes mentioned thus far, as the illuminated (on-state) conductivity is too low to meet the condition that $Z_{PC} < Z_L$, so the optical control of the electric field will not work, and DEP force will not be generated. This does not prevent the use of these ODEP devices with cells, as osmotically balanced low-conductivity buffer solutions can be used. However, the cells can only survive up to several hours in these low-conductivity buffers.

ODEP can be made to work in physiological media if the conductivity of the photosensitive electrode under illumination can be increased. This was accomplished by using an arrayed phototransistor structure as a replacement for the photosensitive electrode (Fig. 4iv), and operation in physiological media was demonstrated (Hsu et al. 2010). However, as with conventional DEP, Joule heating becomes of a concern when operating in physiological media, as the amount of electrical current conducted by the media increases. In the phototransistor-based ODEP device, Joule heating caused a temperature increase, but only by 1–2 °C, and the addition of heat sinks decreased this rise in temperature to less than 1 °C (Pauzauskie et al. 2010).

Unlike optical tweezers, ODEP is not capable of full three-dimensional manipulation; the DEP force decreases with distance from the photosensitive electrode. Thus, the cells under manipulation are always close to a surface, so the nonspecific adherence of cells can be an issue. If cells adhere when it is not desired, their adhesion forces can be on the order of nanonewtons, much larger than typical ODEP forces of tens to hundreds of piconewtons. This nonspecific cell adhesion can be reduced by a factor of 30 by coating the electrodes with a layer of poly(ethylene glycol) (PEG), a common antifouling polymer (Lau et al. 2009). It is also possible to use two photosensitive electrodes and repulsive DEP forces to achieve pseudo-3D manipulation, and to keep the cells under manipulation from contacting the ODEP surfaces (Hwang et al. 2008b). However, the addition of the second photosensitive electrode makes it harder to image the cells, and cells must be constantly manipulated to keep them contacting the electrode surfaces.

3.3 Other Electrokinetic Phenomena

Other electrokinetic forces can be created in a standard ODEP device (Valley et al. 2008). If the applied electric field has a frequency on the order of 100 Hz or less, electrolysis can occur. Bubbles generated by electrolysis can interfere with normal ODEP operation, harm cell membranes, and damage the photosensitive electrode layer, so this operating regime is usually avoided.

As the frequency of the applied electric field increases to a few kHz, light-induced AC electroosmosis (LACE) can result. The optical patterns control the electric field and the zeta potential at the liquid–photoconductor interface, resulting in electroosmotic flow that can be used to transport particles (Chiou et al. 2008; Hwang 2009).

Electrothermal heating can be observed in a standard OET device or similar structures when using high optical intensities or large electric fields. Electrothermal heating can also be exploited to drive optically controllable fluid flows, which can be used to trap and pattern particles (Kumar et al. 2009; Williams et al. 2010; Mishra et al. 2014). Optically controlled electrothermal heating relies on the absorption of light in a substrate to create thermal gradients in a fluid. The thermal gradients result in gradients in the electrical conductivity and permittivity of the fluid, driving electrothermal flow. This flow has been harnessed for the trapping of bacteria and the sorting of microorganisms based on size (Kwon et al. 2012).

4 Optothermal Manipulation

The use of direct optical forces for cell manipulation is limited by the magnitude of forces that can be exerted, which is typically in the piconewton range. However, light can be used to exert larger forces through indirect mechanisms that use the transfer of energy rather than the transfer of photon momentum. Unlike direct optical actuation, photothermal or optothermal actuation methods use light to generate heat through absorption in a material, creating thermal forces.

One optothermal force is thermophoresis, which is the motion of particles due to a temperature gradient. Optothermal heating can also induce thermocapillary effects, which can move bubbles, droplets, and thin liquid films, and can be harnessed to manipulate cells.

4.1 Thermophoresis

Thermophoresis, also known as thermomigration or thermodiffusion, is the response of particles subjected to a temperature gradient, and can be described by

$$U = -D_T \nabla T \tag{4}$$

where U is the drift velocity, D_T is the particle mobility, and T is temperature (Würger 2010). The temperature gradient necessary for thermophoresis can be created by opto-thermal heating, and used for the trapping and transport of biological particles, such as DNA (Duhr and Braun 2006; Weinert and Braun 2009; Maeda et al. 2011), or even the stretching of DNA (Ichikawa et al. 2007), but its use for larger objects such as cells is limited (Helden et al. 2014). However, thermophoresis has been shown to be useful for the manipulation of DNA and other molecules inside living cells (Reichl and Braun 2014), providing a method for the in vivo measurement of intercellular activity.

4.2 Opto-Thermocapillary Cell Manipulation

The thermocapillary or thermal Marangoni effect refers to mass transfer along a liquid–gas interface due to a surface tension gradient created by a temperature gradient. If optical heating creates the temperature gradient that induces the surface tension gradient, the resulting effect can be termed as opto-thermocapillary force. Opto-thermocapillary force has been utilized for actuating various fluidic elements such as bubbles (Ohta et al. 2007d; Tan and Takeuchi 2007; Hu et al. 2011; Rahman et al. 2015a), droplets (Kotz et al. 2004; Farahi et al. 2005; Baroud et al. 2007; Hu and Ohta 2011), and liquid films (Garnier et al. 2003), and can be harnessed for the manipulation of the cells and biomolecules.

Opto-thermocapillary force can be used to actuate microbubbles that are capable of manipulating cells and biomolecules (Hu et al. 2011). Optical patterns are used to create localized hot spots on an absorbing substrate, which are used actuate the microbubbles (Fig. 5). The microbubbles can function as untethered micrometer-scale actuators, and can manipulate various types of cells, including yeast cells (Hu et al. 2012a) and NIH-3T3 fibroblasts (Hu et al. 2013a).

Unlike optical tweezers, opto-thermocapillary actuation is not dependent on the optical properties of the object under manipulation. Opto-thermocapillary actuation is also insensitive to the electrical properties of the liquid medium and the object under manipulation, in contrast to ODEP. However, opto-thermocapillary actuation does share the flexibility of optical control with the other cell manipulation methods described in this chapter, enabling parallel, independent actuation of multiple micro-objects. Although opto-thermocapillary actuation has only been

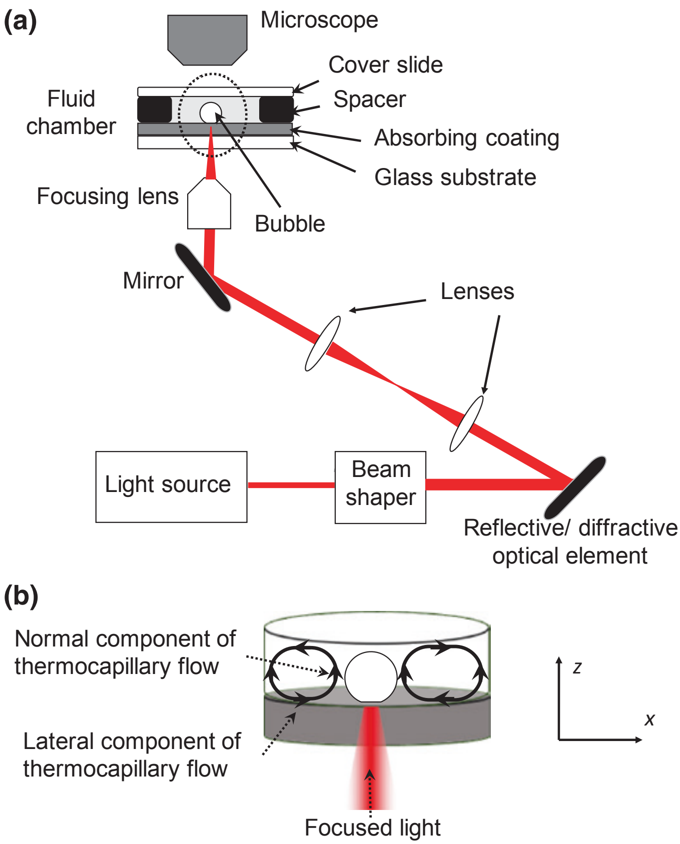

Fig. 5 Opto-thermocapillary bubble manipulation setup. **a** The overall optical setup. The light source is focused on an absorbing coating (typically 1-μm-thick amorphous silicon on 200-nm-thick indium tin oxide) at the floor of a fluidic chamber. **b** Close-up of the portion of the setup circled by the dashed line in **a**, showing the opto-thermocapillary flow surrounding the bubble

demonstrated on up to 12 microbubbles in parallel to date, it has the potential to control at least 50 microbubbles simultaneously (Rahman et al. 2015a).

The microbubbles for opto-thermocapillary actuation are generated in a liquid media by the optothermal heating of the substrate (Fig. 5). A laser beam is focused on the absorptive coating of the substrate, which converts the light energy to thermal energy, causing localized heating in the fluid adjacent to the hot spot (Ohta et al. 2007d). This results in the vaporization of the working fluid at the hot spot, nucleating the bubble.

Once the bubble has been created, the temperature gradient created by the laser-heated region drives a toroidal thermocapillary flow that can be divided into two principal components of interest (Fig. 5b): lateral and normal. The flow that is parallel to the substrate is the lateral part of the thermocapillary flow. Another thermocapillary flow component normal to the substrate is present around the bubble perimeter, due to the temperature gradient in the z-direction. If an object has a density greater than that of the carrier fluid, interaction with the normal flow can be minimized, and the object can be pulled toward the bubble without having the object circulate in the toroidal thermocapillary flow.

The lateral opto-thermocapillary flow was used to pattern yeast cells in an agarose solution. After the yeast cells were patterned, the agarose solution was gelled, and 92 % of the original cells underwent multiple cell divisions over a period of 9 h (Fig. 6) (Hu et al. 2012a). A 980-nm diode laser focused to an intensity of 508 kW/cm^2, pulsed with 60-μs pulses at 60 Hz, was used to generate the bubbles and thermocapillary flow used for manipulation.

Opto-thermocapillary flow has also been used to trap and transport NIH/3T3 fibroblasts (Hu et al. 2013a). The cells were suspended in cell media mixed with a hydrogel or agarose pre-polymer, and patterned using opto-thermocapillary flow in an open reservoir. Fibroblasts in a polyethylene glycol diacrylate (PEGDA) pre-polymer solution were manipulated into a 4 × 4 cell array in 6.5 min (Fig. 7a–d), at a velocity of 30 μm/s. After the assembly, the PEGDA solution was photopolymerized (Fig. 7e), and cell viability was tested (Fig. 7f). Fibroblasts were also assembled into a 3 × 3 cell matrix in a 4 % agarose solution at an average velocity

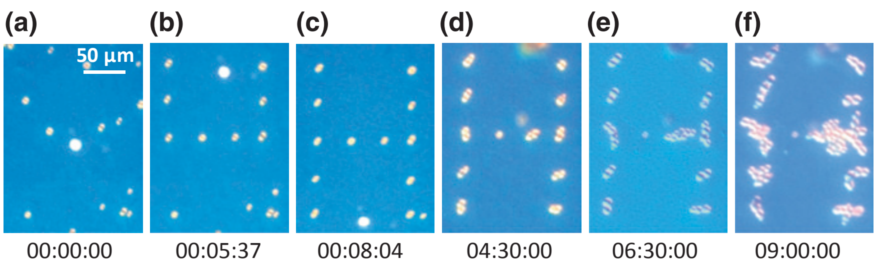

Fig. 6 Yeast cells assembled using opto-thermocapillary flow. **a**–**c** Assembly of yeast cells assembly into an “H” shape in agarose solution. The agarose was subsequently gelled by lowering the solution temperature to 15 °C. **d**–**f** Cell culturing for 9 h after agarose gelation. Reproduced with permission from Hu et al. (2012a)

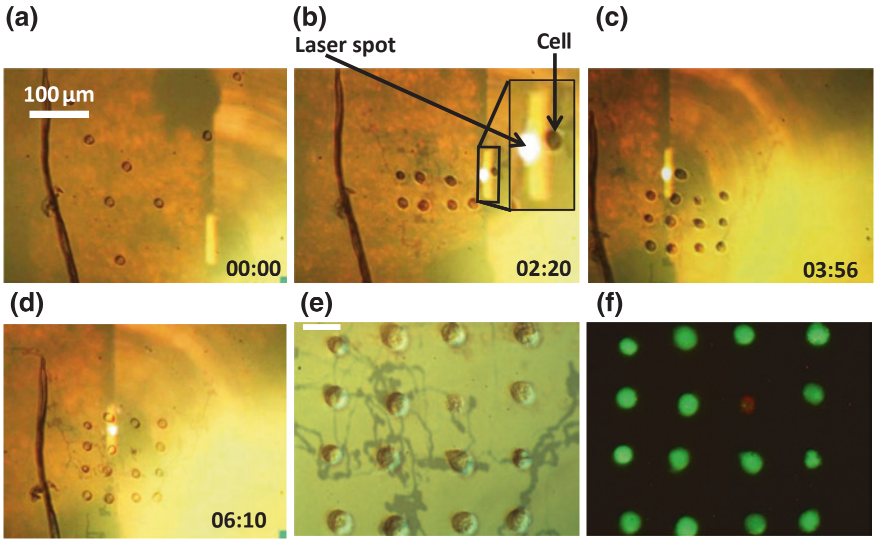

Fig. 7 Opto-thermocapillary assembly of NIH/3T3 fibroblasts. The time stamp format is minutes: seconds. **a–d** Cells are assembled by the opto-thermocapillary flow. The *inset* shows a close-up image of the cell being manipulated. **e** Single-cell pattern after PEGDA gelation. The dark tracks may be due to the laser heating of the substrate, but they do not affect the experiment. The *scale bar* is 20 mm, and also applies to **f**. **f** Green fluorescence indicates viable cells. Only one cell displays red fluorescence, indicating a nonviable cell. Reproduced from Hu et al. (2013a) with permission from the Royal Society of Chemistry

of 8 μm/s. The reduced opto-thermocapillary flow in the more viscous agarose solution resulted in a lower cell manipulation speed, but subsequent agarose gelation and culturing also showed high cell viability.

Additional functionality may be added by using opto-thermocapillary actuated bubbles to move secondary hydrogel structures with features to aid in the assembly of cells (Hu et al. 2012b). Hydrogel structures made of photopolymerized poly(ethylene glycol) diacrylate (PEGDA) were created in various geometries, such as disk-shaped structures with 40–70 μm-radii with a 5-μm-radii cavity at the center. A bubble in this central cavity is opto-thermally actuated, actuating the hydrogel structure. This enables the assembly of cells in an uncovered fluid reservoir, providing easy access to the cells under manipulation for downstream culturing or assays. The hydrogel structures can also be made to absorb the incident laser light by adding carbon particles (Hu et al. 2013b). This means that the opto-thermocapillary actuation of the hydrogel structures does not require a light-absorbing substrate.

To achieve the parallel manipulation of the microbubbles using opto-thermocapillary actuation, optical patterns are created from a single laser by a scanning mirror (Hu et al. 2014) or a spatial light modulator (Rahman et al. 2015a). Both systems are similar to methods used to create multiple optical tweezers traps.

Using orthogonal scanning mirrors, a single beam of light can be directed to multiple locations on a two-dimensional plane, controlling up to 10 microbubbles independently at once (Hu et al. 2014). The maximum number of microbubbles was limited by the manual control system, but the time-sharing of one laser beam fundamentally limits the maximum number of microbubbles, just as with time-sharing optical tweezers systems.

Thus, as with holographic optical tweezers, an alternate method is the use of an SLM to create a holographic control system (Rahman et al. 2015a). Instead of redirecting a single beam multiple times, an SLM is able to split the beam using a computer-generated hologram, eliminating the need for time-sharing, and allowing the bubbles to be continuously actuated on a sample.

The control system that was used to realize the control of multiple microbubbles using an SLM was a modified version of the Red Tweezers program (Bowman et al. 2014). Some modifications to the program were necessary, including a bubble-collision-detection system and an error detection system. In addition, an automatic sequence generator was added that allows the microbubbles to move in independent trajectories at the same time. A user defines the path of microbubbles by planning out a sequence of holograms, which is then played on the SLM at specific time intervals for the purpose of particle manipulation (Fig. 8).

Like other cellular manipulation systems, opto-thermocapillary flow has its limitations, some of which are shared with HOT. If a holographic control system is

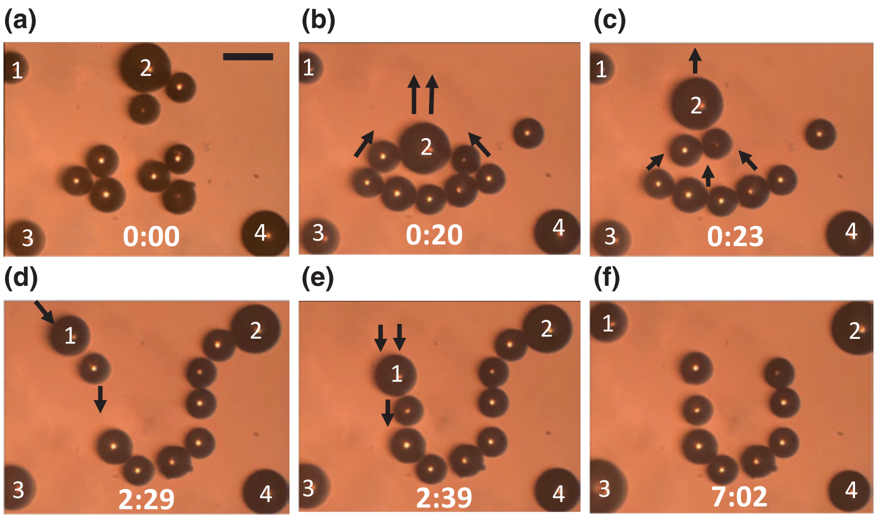

Fig. 8 Assembly of glass beads using multiple opto-thermocapillary flow-actuated microbubbles. Two microbubbles were used to assemble eight glass beads. Each microbubble in the field-of-view is numbered (a total of four); the *scale bar* is 200 µm. **a** Initial position of the microbubbles and the beads. **b**, **c** Bubble 2 pulling the beads into place. **d**, **e** Bubble 1 pushing the beads into place. **f** The final U-shaped pattern, after a total assembly time of 7 min. Reproduced with permission from Rahman et al. (2015b)

used, “ghost traps” may occur, just as in HOT. This can be resolved by using the same iterative algorithms used for HOT systems, resulting in the reduction of aberrations in the system.

In addition, the number of bubbles available for micromanipulation is limited by the available laser power and the working area. Currently, micromanipulation occurs over an area of 1.1 mm by 0.83 mm, and is limited by the magnification of the focusing lens (Rahman et al. 2015b). A smaller magnification provides a larger area of manipulation, but reduces the resolution, as this also increases the minimum size of the optical features that can be created. Therefore, there is a tradeoff between working area and resolution. This could be addressed by using a high-magnification focusing lens for higher resolution, and moving the fluid chamber and microrobots independently to cover a large working area; however, this will increase the complexity of the control system.

Like ODEP, opto-thermocapillary manipulation is currently restricted to two-dimensional motion. However, the potential for three-dimensional opto-thermocapillary manipulation exists, especially if the normal thermocapillary flow can be exploited.

5 Optical Manipulation for Single-Cell Analysis

The optical manipulation methods described in this chapter are useful for enabling or augmenting functions related to single-cell analysis, including cell culturing, cell separation and sorting, cell trapping, isolation, and transport, molecular delivery to cells, cell lysis, and measurements of cellular properties. Specific examples that realize these functions will be described in this section.

5.1 Cell Culturing

The first step in single-cell analysis is the culturing of cells. Optical manipulation can assist with this by enabling the patterning of cells to study cell–cell interactions, or to build cocultures of different types of cells. Traditional cell culture is performed on two-dimensional surfaces, such as tissue culture polystyrene (TCPS). However, cells can be grown on other surfaces, including glass and silicon. Thus, optical manipulation can be used to position cells before subsequent culturing. This has been demonstrated with optical tweezers (Townes-Anderson et al. 1998), phototransistor-based ODEP (Hsu et al. 2010), and optothermal manipulation (Hu et al. 2013a). Although standard ODEP devices do not work in cell culture media, microfluidic structures can be added to automatically exchange low-conductivity ODEP media with cell culture media (Lee et al. 2014), so cell culturing with these types of structures can be possible.

While cell culturing on a two-dimensional planar surface is widespread, more accurate models of cell behavior can be obtained by culturing cells in three dimension, which more closely approximates an in vivo environment. This is typically done by culturing cells in some type of scaffold material made of natural extracellular matrix materials or synthetic materials such as hydrogels. Cells embedded in microscale scaffolds form “microgels” (Yeh et al. 2006) which can be used as building blocks of in vitro tissues and organ-like structures. Optical manipulation can be used to position and organize these microgels, by direct optical manipulation (Kirkham et al. 2015), ODEP (Lin et al. 2012a), or optothermal manipulation (Hu et al. 2012b). Optical manipulation can also assist with the transport and release of chemicals to control cell growth and differentiation during culturing (Kirkham et al. 2015).

5.2 Cell Sorting

Once cells are cultured, cells of interest need to be separated from unwanted cells through some form of cell sorting. This can also be the first step in a single-cell analysis process that uses primary cells, which undergo minimal, if any, culturing. Some examples of sorting that are interesting include sorting by the type of cell, such as red blood cells from white blood cells, or cancerous cells from healthy cells. Sorting of live cells from dead cells can also help with downstream processes, as well as the isolation of rare cells from a larger cell population.

Various mechanisms can be used to obtain these sorting results (Fig. 9), and can broadly be classified as passive sorters, which use static optical patterns, or active sorters, which use dynamic optical patterns. Cell types can be sorted by size, using various mechanisms including direct optical force or ODEP. Direct optical forces can also distinguish cells based upon their optical properties, while ODEP can be used to differentiate cells based on their electrical properties. Importantly, sorting upon these intrinsic properties means that labeling is not required, minimizing sample preparation and adverse effects on the cells to be sorted. All types of optical manipulation systems can also sort cells based upon visual characteristics identified by image-based control systems, such as morphology or fluorescent markers.

5.2.1 Cell Sorting with Direct Optical Forces

The forces resulting from optical landscapes can enable sorting through the variation in the strength of particle interactions. A three-dimensional optical lattice produced by multi-beam interferences creates a landscape in which particles can be separated by size and refractive index (Fig. 9ii) (MacDonald et al. 2003). These variations in force can be used to separate cells that are otherwise indistinguishable. Moving optical interference patterns can also enable size- and shape-based separation of cells (Jákl et al. 2014).

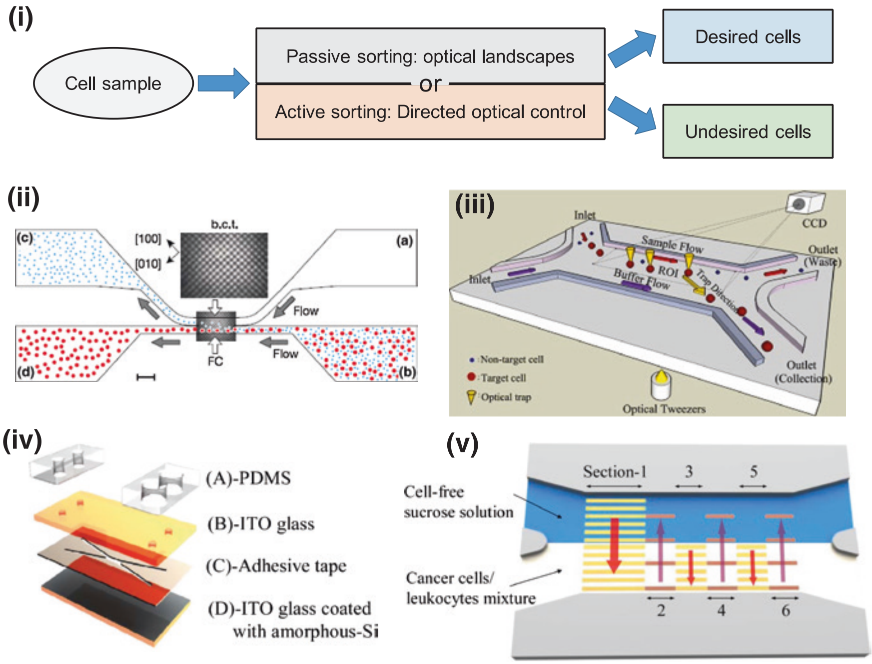

Fig. 9 Optical cell sorting. **i** Block diagram of an optical cell sorter. A cell sample is sorted by either a passive (static) optical landscape, or by an active (dynamic) force applied to the cells. **ii** An example of a passive cell sorter, in which an optical lattice is used to sort objects based on their size or index of refraction. Reproduced with permission from MacDonald et al. (2003). **iii** An active cell sorter that uses multiple optical tweezers traps to move desired cells into an outlet channel. Reproduced from Wang et al. (2011) with permission from the Royal Society of Chemistry. **iv** An exploded view of active cell sorting device that uses ODEP. **v** The operation of the ODEP-based active cell sorter, showing dynamic light patterns that are used to sort the cells by cell type. **iv** and **v** are reproduced from Huang et al. (2013c) with permission from the Royal Society of Chemistry

Optical tweezers can provide the force needed to sort cells in a microfluidic device. Many of these systems have automatic image-based sorting that allows cells to be sorted by parameters including size, shape, and fluorescence. For example, image processing was used to identify the size of cells, triggering optical tweezers to move particular cells into a collection channel, achieving the sorting of 3- to 5-μm-diameter yeast cells from 10-μm-diameter polystyrene beads at a yield of 100 % and a recovery rate of 90 % (Lin et al. 2008). Cells can also be sorted based on more than one parameter. One such microfluidic sorter is able to sort cells by size, shape, and fluorescence (Landenberger et al. 2012). Using this system, 2-μm-diameter yeast cells were sorted with a throughput of 1.4 cells per second and a recovery rate of 95 % using fluorescence, size, and a combination of fluorescence and size.

HOT systems can also be integrated with microfluidic systems for automatic cell sorting (Fig. 9iii). Using HOT to generate multiple optical traps can also result in a higher sorting throughput. An HOT-enabled automatic cell sorter is able to sort particles by both size and fluorescence, as demonstrated by the sorting of 5- to 8-μm-diameter yeast from 2-μm-diameter polystyrene micro-beads, and 10- to 15-μm-diameter fluorescently tagged stem cells from a mixture other similarly sized cells with a 90 % recovery rate and 90 % purity (Wang et al. 2011).

Cell sorters can also utilize the optical scattering force, which can be used to extract cells out of microwells and transported downstream for further analysis or disposal (Kovac and Voldman 2007). This image-based sorter has a large workspace for increased sorting throughput, although cells do need to be fluorescently tagged. However, the intrinsic response of cells to an optical scattering force can also be used for label-free sorting. This was demonstrated in a continuous-flow microfluidic sorter of red blood cells, lymphocytes, granulocytes, and monocytes (Lee et al. 2013). Each of these blood cells has different scattering forces, allowing each type of cell to be successfully distinguished from each other.

5.2.2 Cell Sorting with ODEP

Different cell types may also have different electrical properties, such as permittivity and conductivity, which can be distinguished using ODEP. Sorting based upon electrical properties can enable sorting that is not possible solely based on physical size, such as the separation of live cells from dead cells (Chiou et al. 2005). A live cell can maintain an ion differential across its interior and the surrounding medium. If the cell is in a low-conductivity medium, as in the case for standard ODEP manipulation, the interior of the cell will have an electrical conductivity greater than the medium. If the cell dies, the ion differential is no longer maintained, and the conductivity of the cell interior becomes similar to that of the low-conductivity medium. This makes the ODEP force exerted on live and dead cells different, according to Eqs. 1–3, and can even result in a change in the direction of the ODEP force. For instance, live white blood cells were observed to have a positive ODEP force, while dead cells had a negative ODEP force. This difference in force was used to selectively concentrate the live cells (Chiou et al. 2005). A similar observation with sperm cells enabled the discrimination between live and dead spermatozoa, even among completely nonmotile cells (Ohta et al. 2010). Variations in the electrical properties of different types of cells can also be exploited by ODEP. Physiological differences between B lymphocytes and red blood cells resulted in a difference in DEP force (Zhao et al. 2013), as well as for HeLa and Jurkat cells (Ohta et al. 2007b), prostate cancer cells and leukocytes (Huang et al. 2012, 2013c), and oral cancer cells and leukocytes (Huang et al. 2013c). These separations can be used to realize constant-flow cell sorting; up to 80 % cell purity and 80 % cell recovery with 95 % cell viability was achieved using such a device (Fig. 9iv, v) (Huang et al. 2013c).

Visual characteristics can also be an important sorting trigger for ODEP-based sorting. This can include checking for different cell morphologies and features, or by checking for visual tags, such as fluorescent markers. For example, T-cells and dendritic cells were distinguished via fluorescent tags, and ODEP was used to separate the different cell types (Jeorrett et al. 2014). The tags provided visual differentiation between the cells, enabling sorting; this separation was not possible based solely on electrical properties, as the two types of cells experienced similar ODEP forces.

5.3 Cell Trapping and Isolation

Once cells of interest are sorted, single cells need to be isolated for analysis. All of the optical trapping techniques discussed here are capable of this task, as described earlier in this chapter. The trapping of cells is usually done in vitro, which is suitable for single-cell analysis procedures. However, for some applications, it may be desirable to trap cells in vivo. Optical manipulation in vivo is more challenging, due to optical absorption by tissues, and inhomogeneous indices of refraction, but optical tweezers have been used to manipulate red blood cells in vivo, in the bloodstream of living mice (Zhong et al. 2013). The trapped red blood cells were manipulated in three dimensions, and the optical trap was shown to be capable of causing and clearing artificial clots in capillaries by transporting the blood cells to the appropriate locations. The stiffness of these optical traps was approximately 10 pN/μm with a power of 168 mW at the trap focus. Importantly, no thermal damage was observed on the cells that were trapped or the tissues in the path of the laser, showing the promise of optical tweezers for in vivo cell manipulation.

5.4 Cell Surgery and Lysis

Further analysis of single cells is necessary to enable a deeper understanding of cellular behavior. This type of analysis may require access to the cell interior, which can be achieved by using lasers to cut into cell membranes (Berns 1998). Organelles can then be extracted from the cell (Shelby et al. 2005), or cargo can be delivered to the cell. Lasers can also be used to realize cell fusion (Steubing et al. 1991) and the attachment of cells to one another without fusion (Katchinskiy et al. 2014). Cell lysis can also be induced through a more indirect process in which pulsed lasers are used to create cavitation bubbles that disrupt the cell membrane (Rau et al. 2004; Lai et al. 2008). Single-cell resolution is possible if cells are confined in microchannels (Lai et al. 2008), or in picoliter or femtoliter droplets (He et al. 2005). Alternatively, cell lysis can also be done using ODEP

(Lin and Lee 2009; Witte et al. 2014), including selective lysis depending upon the cell's shape (Kremer et al. 2014).

A related method of optically controlled cell surgery is the photothermal nanoblade, which utilizes titanium-coated microcapillaries (Wu et al. 2011) or vias (Wu et al. 2015) to convert nanosecond laser pulses to localized cavitation bubbles. These bubbles create pores in cell membranes, allowing cargo to be delivered into the cells. The cargo can cover a very large size range: 1 nm (RNA) up to 2 μm (bacteria) (Wu et al. 2011), and a throughput of up to 100,000 cells per minute can be achieved (Wu et al. 2015). This method can be nondestructive and used to deliver payloads to cells, like the methods that are covered next, in Sect. 5.5.

5.5 Molecular Delivery to Cells

A gentler operation than cell lysis is the creation of pores in the cell membrane, which can be reversible, where the pores self-heal, or irreversible, where the pores remain. Cell lysis, previously covered in Sect. 5.4, is irreversible membrane poration taken to the extreme. However, if the membrane pores are small enough, cells can recover from this operation, enabling molecular delivery. This enables techniques such as gene therapy or cell transformation. ODEP can use electric field pulses to open pores in the membrane of cells, enabling molecular delivery. In addition, the light patterns used by ODEP can be used to select specific cells for the membrane poration operation, while leaving other cells unaffected (Valley et al. 2009; Wang et al. 2014).

Opto-thermal heating of the cell membrane by a continuous-wave laser, facilitated by light-absorbing dyes, can temporarily increase membrane permeability, enabling molecular delivery (Palumbo et al. 1996; Schneckenburger et al. 2002; Nikolskaya et al. 2006). Single-cell resolution is possible with this method, although the transfection efficiency is relatively low (less than 30 %) (Stevenson et al. 2010).

Another method of selective cell poration is opto-thermocapillary size-oscillating microbubbles (Fan et al. 2014, 2015). Microsecond laser pulses focused on a light-absorbing surface nucleate a vapor microbubble. The size of the bubble is controlled by the laser pulses, causing the bubble to expand and collapse. This oscillation in the size of the bubble produces a fluid flow that can induce transient pores in a nearby cell membrane (Fig. 10i). This method has single-cell resolution, and can reach up to 95 % of molecular delivery efficiency while maintaining 95 % cell viability. Multiple molecules have been delivered to the same cell sample using this method (Fig. 10ii).

Nanosecond-pulsed lasers create shockwaves by laser-induced cavitation bubble expansion, which can open pores in cell membranes (Tsukakoshi et al. 1984; Soughayer et al. 2000; Venugopalan et al. 2002; Hellman et al. 2008; Wang et al. 2010b). This mechanism has a relatively low resolution; the shockwaves can affect cells up to 60 μm away from the laser focal point (Soughayer et al. 2000).

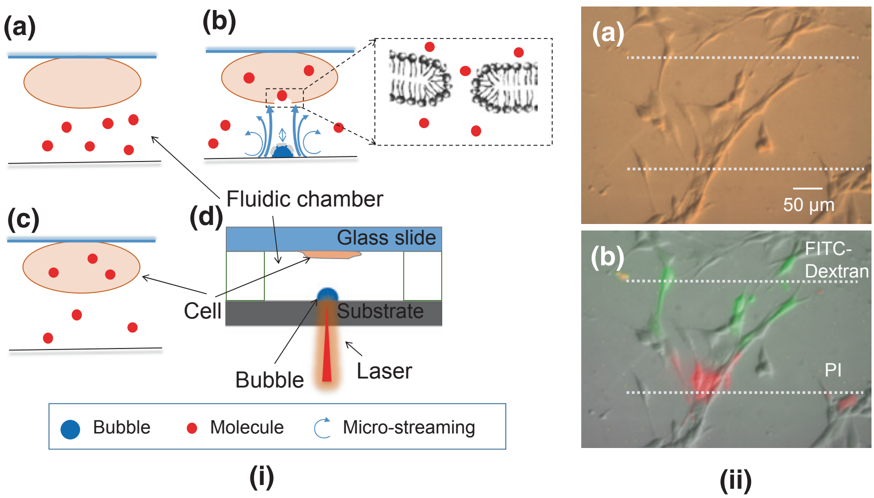

Fig. 10 Molecular delivery by laser-induced microbubbles. **i** The molecular delivery process. *(i-a)* The cell membrane is initially impermeable to the molecules. *(i-b)* A size-oscillating bubble is nucleated below the target cell by laser heating, creating shear stress that opens transient pores in the cell membrane, allowing the molecules to enter the cell interior. *(i-c)* The laser excitation ceases, and the cell membrane heals. *(i-d)* Cross section of the fluidic chamber. The laser is absorbed by the substrate, nucleating the bubble. **ii** Targeted delivery of multiple molecules. *(ii-a)* Bright-field image of fibroblasts. Molecular delivery of 3-kDa FITC-Dextran dye is performed on the cells in the vicinity of the *upper dashed line*. Propidium iodide (PI) is delivered to the cells marked by the *lower dashed line*. *(ii-b)* Composite image showing successful molecular delivery of FITC-Dextran (green fluorescence) and PI (red fluorescence). Reproduced from Fan et al. (2015) with permission from the Royal Society of Chemistry

However, single-cell resolution can still be achieved by nanosecond-pulsed lasers with the assistance of microfluidic structures to localize and confine cells (Le Gac et al. 2007; Li et al. 2013).

Femtosecond-pulsed lasers can also open transient pores in cell membranes via multiphoton ionization and plasma formation (Vogel et al. 2005). Transfection efficiencies using femtosecond-pulsed lasers are greater than 90 % for various cell lines (Tirlapur and Konig 2002; Baumgart et al. 2008; Uchugonova et al. 2008), and slightly lower (around 70 %) for stem cells and primary cells (Baumgart et al. 2008; Uchugonova et al. 2008). However, submicron focusing accuracy on the cell membrane is necessary for two-photon absorption to occur; a focusing error of 3 μm can decrease transfection efficiency by more than 50 % (Tsampoula et al. 2007). This focusing issue can be mitigated by using microparticles trapped by optical tweezers to focus the femtosecond laser precisely on the cell membrane (Waleed et al. 2013), or by using a Bessel beam (Tsampoula et al. 2007; Cižmár et al. 2008). In vivo transfection with femtosecond lasers has also been demonstrated (Zeira et al. 2007). The main barrier to the wide deployment of femtosecond-pulsed laser systems is the high cost of the laser and other equipment, which is on the order of $100,000.

5.6 Measuring Cellular Properties

The final step in a single-cell analysis procedure is to do measurements on the single cells of interest. This is primarily done with some form of microscopy, which is well suited to integration with optical systems, as most optical manipulations are built around or integrated into microscopes. Optical microscopy and sensing will not be reviewed here, but some measurement techniques unique to optical manipulation should be mentioned. For example, the mechanical properties of cells can be measured using optical manipulation (Fig. 11). Optical tweezers can be used to stretch cells, which provides information about the various cell moduli (Lim et al. 2004). Similarly, HOT can be used to measure the dynamic mechanical response of stem cells, from which information such as Young's modulus and viscosity can be extracted (Tan et al. 2012). This can be used to distinguish between differentiated and undifferentiated stem cells.

Optical tweezers can also help with the measurement of the Raman spectrum of cells, which is useful for determining their chemical composition and structure. Optical tweezers are a natural fit for Raman spectroscopy, as the optical trap stabilizes the particle under observation, even if it is in suspension, allowing cells to be accurately measured (Ajito and Torimitsu 2002). The focused laser beam used to create the optical trap can also be used to excite the particle for Raman measurements, or a separate laser can be used. Optical tweezers integrated with Raman spectroscopy has successfully measured various characteristics of cells, including distinguishing between live and dead cells (Xie et al. 2002), changes

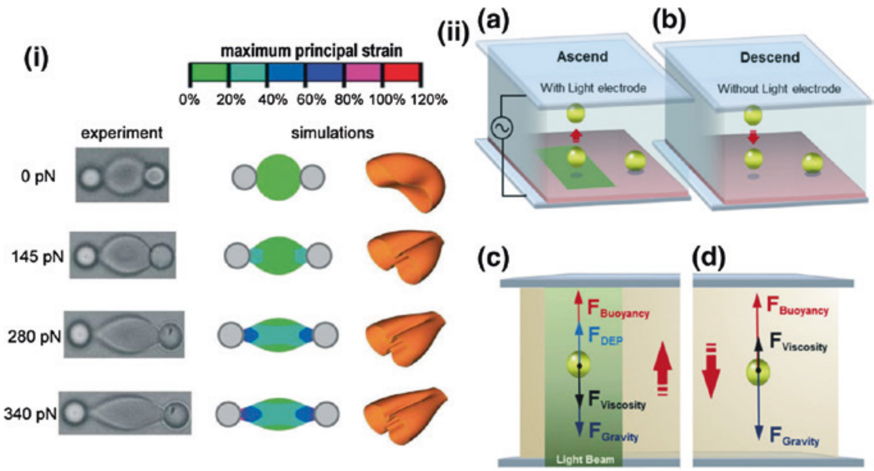

Fig. 11 Measuring cell properties using optical manipulation. **i** Optical tweezers can be used to stretch cells in order to gain information on the cell's mechanical properties. Reproduced from Lim et al. (2004) with permission. **ii** Measuring the mass and density of single cells by levitating the cells using ODEP, then observing the descent of the cells. Reproduced from Zhao et al. (2014) with permission from the Royal Society of Chemistry

in heat-treated cells (Xie et al. 2003), and the oxygenation capability of healthy and diseased red blood cells (De Luca et al. 2008). By adding optical tweezers, Raman spectroscopy and cell sorting can be achieved in the same platform (Lau et al. 2008).

Optical tweezers can be useful for the positioning of cells undergoing measurements. Synchrotron-radiation-based X-ray fluorescence (XRF) microscopy can provide spatial information on the material composition of cells, but this measurement requires that the sample remain still for up to hours at a time. Thus, cells measured using XRF need to be fixed in place, but traditional methods of immobilizing the cells can affect the measurement. To address this, an XRF system was integrated with a HOT system, allowing the cells to be immobilized with minimal effects on the XRF measurement (Vergucht et al. 2015). This system allows the cell under observation to be kept in its natural aqueous environment, reducing sample preparation, and increasing the biological relevance of the results.

HOT systems can collect more complex data such as the three-dimensional refractive index of a cell, which provides information about the composition and the structure of the cell (Habaza et al. 2015). Using HOT, cells could be captured and rotated 180° while in suspension. This is an easier and more efficient way to collect the three-dimensional refractive index data since a wide angular range can be achieved by HOT manipulation without an external rotation stage for the sample.

Cell stiffness is another important mechanical property, as this is an indicator of cellular behavior and cellular health. The relative stiffness of red blood cells can be measured using ODEP, by observing the change in the diameter of the cells when the electric field is applied (Neale et al. 2012). ODEP can also be used to measure the mass of single cells (Zhao et al. 2014, 2015). This method uses ODEP force to levitate cells; subsequently, the ODEP force is removed, and the rate of the descent of the cells is measured. This allows calculation of the density and mass of the cells (Fig. 11ii). ODEP force can also be used to detect the health of a cell, such as oocytes cultured under normal conditions compared to oocytes kept in a nutrient-free solution for 3 days (Hwang et al. 2009). A similar principle was used to differentiate between healthy and unhealthy embryos (Valley et al. 2010).

6 Summary

Optical manipulation encompasses a variety of technologies that are useful for single-cell analysis. Compared to other micromanipulation techniques, optical manipulation is usually the most flexible and adaptable, as reconfiguration of the manipulators can be done by altering the optical pattern. It is even possible to create reconfigurable microfluidic channels using optical control (Haulot et al. 2012), making it possible to realize a complete, optically controlled single-cell analysis platform.

Optical manipulation is especially suitable as an adaptable tool that can be reconfigured for different applications. It also excels at the manipulation of *specific* single cells, as appropriate optical patterns can be generated on the fly to target cells of interest.

However, optical manipulation usually requires expensive or complicated equipment, or both. Thus, it begins to lose some attractiveness for highly repetitive tasks that do not take advantage of the dynamic, flexible nature of optical control. In addition, if the manipulation of specific cells is not required, optical manipulation is less favorable, as it tends to have lower throughput than batch processes.

Optical manipulation has already proven useful in the analysis of single cells, and there is much potential for its continued use and expansion to other applications, especially if the inherent advantages of optical manipulation are exploited.

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